

EXPERIMENT – 01

Objective : Write a program of “hello world” in C++.

Source Code :

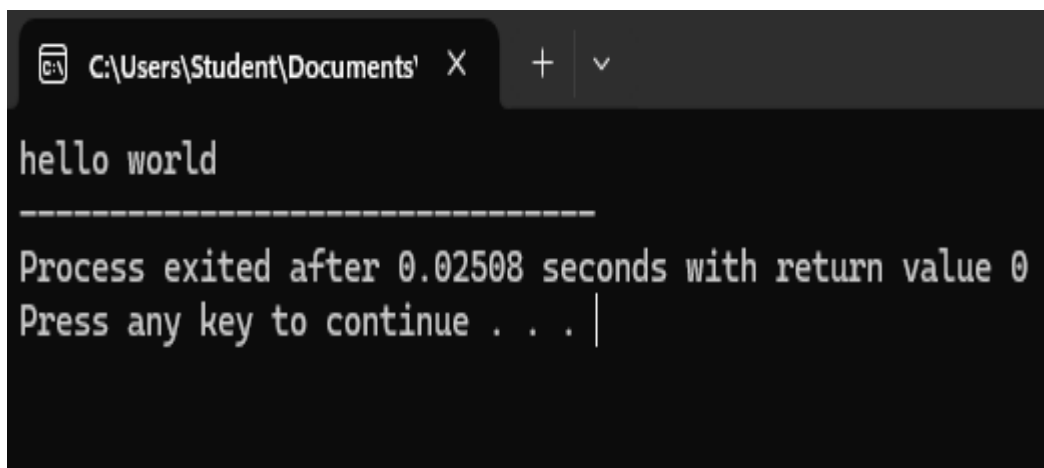
```
#include<iostream>

using namespace std;

int main()
{
    cout<<"hello world";

    return 0;
}
```

Output :

A screenshot of a Windows command prompt window. The title bar shows the file path 'C:\Users\Student\Documents' and standard window controls. The command prompt displays the output of a C++ program: 'hello world' followed by a dashed line separator. Below the separator, it shows 'Process exited after 0.02508 seconds with return value 0' and 'Press any key to continue . . . |' with a cursor at the end.

```
C:\Users\Student\Documents' X + v

hello world
-----
Process exited after 0.02508 seconds with return value 0
Press any key to continue . . . |
```

EXPERIMENT – 02

Objective : Write a program to perform arithmetic operations in C++.

Source code :

```
#include<iostream>

using namespace std;

int main()
{
    int a = 10, b = 30, sum, sub, multi, div;

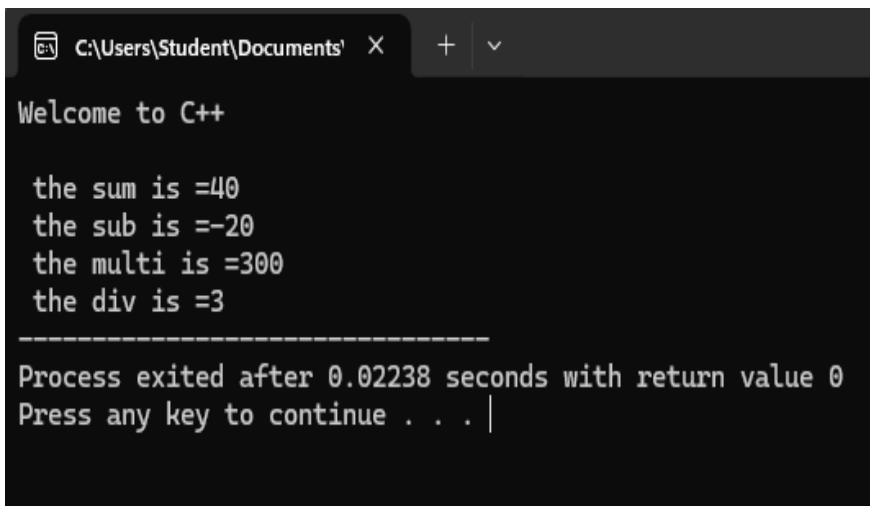
    cout<<"Welcome to C++\n";

    sum = a + b;
    sub = a - b;
    multi = a * b;
    div = b / a;

    cout<<"\n the sum is =" <<sum;
    cout<<"\n the sub is =" <<sub;
    cout<<"\n the multi is ="<<multi;
    cout<<"\n the div is ="<<div;

    return 0;
}
```

Output :

A screenshot of a Windows command prompt window. The title bar shows the file path 'C:\Users\Student\Documents' and standard window controls. The output of the C++ program is displayed in a monospaced font. It starts with 'Welcome to C++' on a new line. Then, it shows the results of arithmetic operations: 'the sum is =40', 'the sub is =-20', 'the multi is =300', and 'the div is =3', each on a new line. A horizontal dashed line separates the output from the final status message: 'Process exited after 0.02238 seconds with return value 0'. Below this, it says 'Press any key to continue . . . |' with a cursor at the end.

```
C:\Users\Student\Documents' X + v

Welcome to C++

the sum is =40
the sub is =-20
the multi is =300
the div is =3
-----
Process exited after 0.02238 seconds with return value 0
Press any key to continue . . . |
```

EXPERIMENT –03

Objective : Write a program to find the biggest no. between three numbers.

Source code:

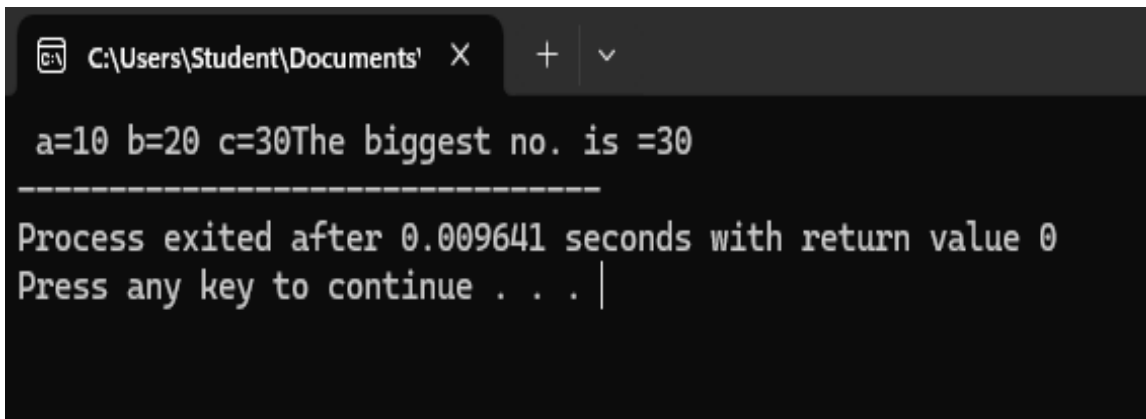
```
#include<iostream>

using namespace std;

int main()
{
    int a = 10, b= 20, c = 30;
    cout<<" a="<<a<<" b="<<b<<" c="<<c;
    if(a>b && a>c)
    {
        cout<<"The biggest no. is "<<a;
    }
    else if(b>a && b>c) {
        cout<<"The biggest no. is "<<b;
    }
    else
        cout<<"The biggest no. is "<<c;

    return 0;
}
```

Output :



```
C:\Users\Student\Documents X + v
a=10 b=20 c=30The biggest no. is =30
-----
Process exited after 0.009641 seconds with return value 0
Press any key to continue . . . |
```

EXPERIMENT – 04

Objective : Write a program in C++ to swap numbers without variable.

Source Code :

```
#include<iostream>

using namespace std;

int main ()
{
    int a = 11, b = 22, temp;

    cout<<"The initial value of a & b"<<endl;

    cout<<"a ="<<a<<endl;

    cout<<"b ="<<b<<endl;

    temp = a;

    a = b;

    b = temp;

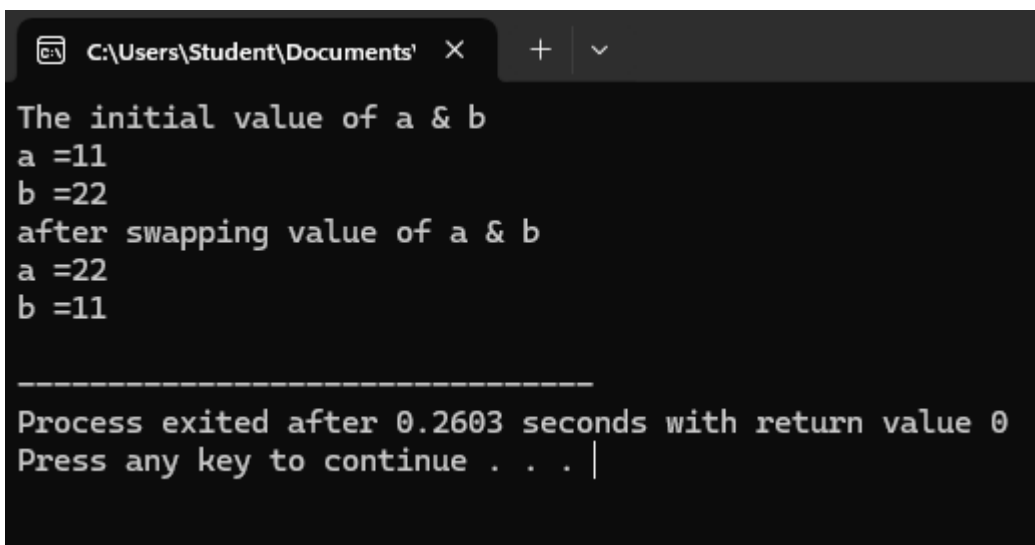
    cout<<"after swapping value of a & b"<<endl;

    cout<<"a ="<<a<<endl;

    cout<<"b ="<<b<<endl;

    return 0;
}
```

Output :

A screenshot of a Windows command prompt window. The title bar shows the file path 'C:\Users\Student\Documents\' and a close button. The terminal output displays the results of the C++ program: 'The initial value of a & b', 'a =11', 'b =22', 'after swapping value of a & b', 'a =22', and 'b =11'. Below this, a separator line is followed by the message 'Process exited after 0.2603 seconds with return value 0' and 'Press any key to continue . . . |' with a cursor.

```
C:\Users\Student\Documents' X + v
The initial value of a & b
a =11
b =22
after swapping value of a & b
a =22
b =11

-----
Process exited after 0.2603 seconds with return value 0
Press any key to continue . . . |
```

EXPERIMENT – 05

Objective : Write a program in C++ using swap number with variable.

Source code:

```
#include<iostream>

using namespace std;

int main()
{
    int a = 11, b = 22;

    cout<<"The initial value of a & b"<<endl;

    cout<<"a ="<<a<<endl;

    cout<<"b ="<<b<<endl;

    a = a + b;

    b = a - b;

    a = a - b;

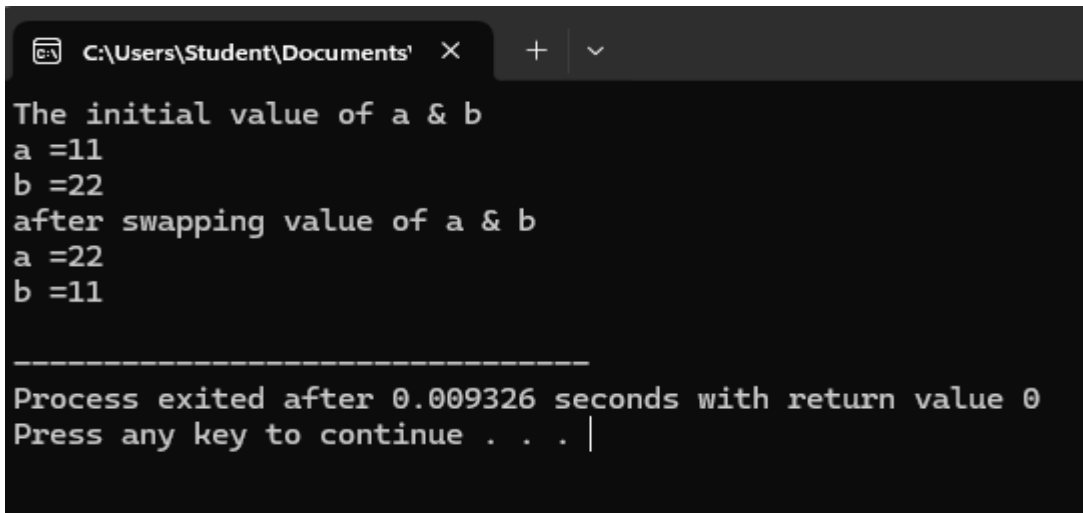
    cout<<"after swapping value of a & b"<<endl;

    cout<<"a ="<<a<<endl;

    cout<<"b ="<<b<<endl;

    return 0;
}
```

Output :

A screenshot of a Windows command prompt window. The title bar shows the file path 'C:\Users\Student\Documents' and standard window controls. The terminal output displays the results of the C++ program: 'The initial value of a & b', 'a =11', 'b =22', 'after swapping value of a & b', 'a =22', and 'b =11'. Below this, a separator line is followed by the message 'Process exited after 0.009326 seconds with return value 0' and 'Press any key to continue . . . |' with a cursor.

```
C:\Users\Student\Documents' X + v

The initial value of a & b
a =11
b =22
after swapping value of a & b
a =22
b =11

-----
Process exited after 0.009326 seconds with return value 0
Press any key to continue . . . |
```

EXPERIMENT – 06

Objective : Write a program to perform arithmetic operation with user input.

Source code :

```
#include<iostream>

using namespace std;

int main()
{
    int a, b, sum, sub, multi, div;

    cout<<"Enter the value of a =";

    cin>>a;

    cout<<"Enter the value of b =";

    cin>>b;

    sum = a + b;

    sub = a - b;

    multi = a * b;

    div = b / a;

    cout<<"\n the sum is =" <<sum<<endl;

    cout<<"\n the sub is =" <<sub<<endl;

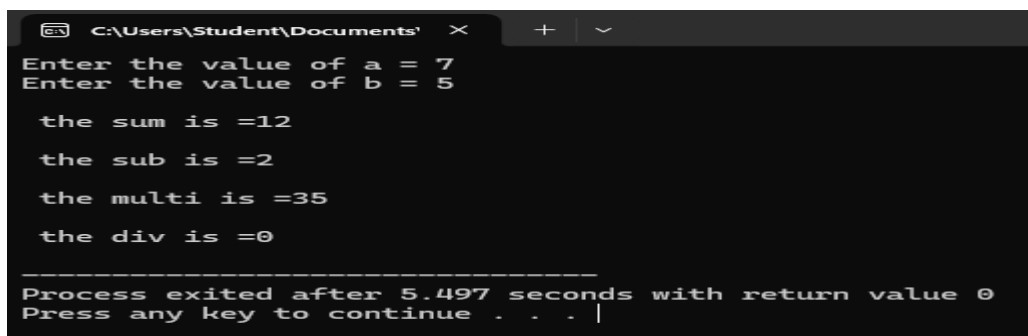
    cout<<"\n the multi is ="<<multi<<endl;

    cout<<"\n the div is ="<<div<<endl;

    return 0;

}
```

Output :

A screenshot of a Windows command prompt window showing the execution of a C++ program. The window title is 'C:\Users\Student\Documents'. The program prompts for two integers, 'a' and 'b'. The user enters '7' for 'a' and '5' for 'b'. The program then calculates and displays the sum (12), subtraction (2), multiplication (35), and division (0) of these two numbers. At the bottom, it shows the process exiting after 5.497 seconds with a return value of 0, and prompts the user to press any key to continue.

```
C:\Users\Student\Documents' x + v
Enter the value of a = 7
Enter the value of b = 5

the sum is =12
the sub is =2
the multi is =35
the div is =0

-----
Process exited after 5.497 seconds with return value 0
Press any key to continue . . . |
```

EXPERIMENT – 07

Objective : Write a program to perform factorial of any number.

Source code :

```
#include<iostream>

using namespace std;

int main()
{
    int i, num, fact = 1;

    cout<<"Enter any number: ";

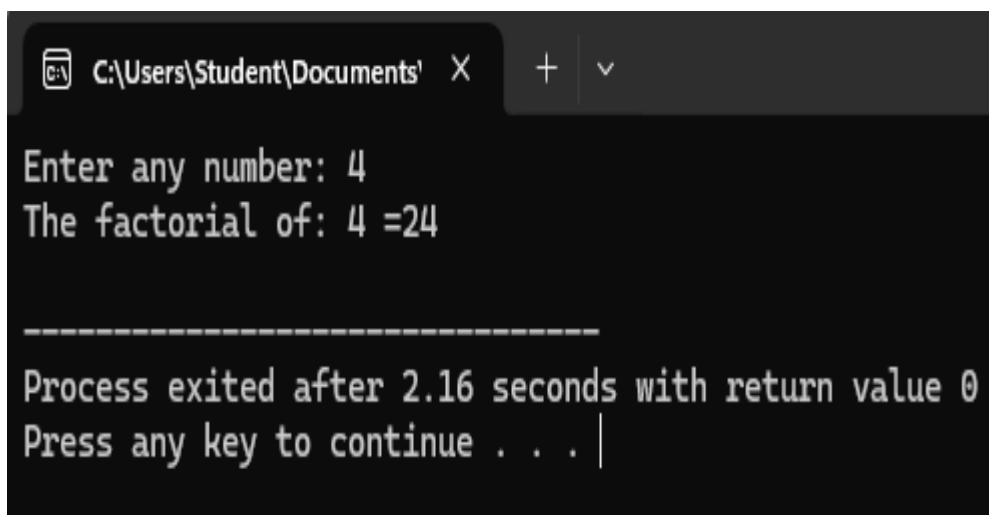
    cin>>num;

    for(i=1; i <= num; i++) {
        fact = fact*i;
    }

    cout<<"The factorial of: "<<num<<" = "<<fact<<endl;

    return 0;
}
```

Output :

A screenshot of a Windows command prompt window. The title bar shows the file path 'C:\Users\Student\Documents' and standard window controls. The command prompt displays the following text: 'Enter any number: 4', 'The factorial of: 4 =24', a dashed line separator, and 'Process exited after 2.16 seconds with return value 0'. The prompt 'Press any key to continue . . . |' is visible at the bottom, with a vertical cursor line.

EXPERIMENT – 08

Objective : Write a program to perform table of any number.

Source code :

```
#include<iostream>

using namespace std;

int main()
{
    int i, num ,n;

    cout<<"Enter any number: ";

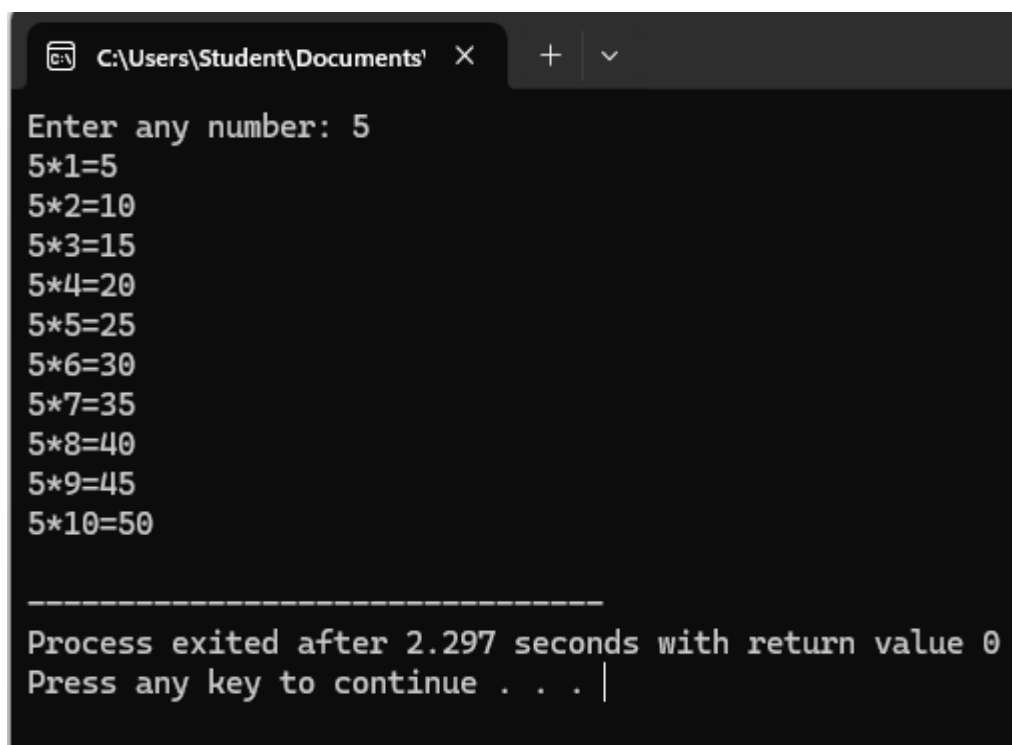
    cin>>num;

    for(i=1; i<=10; i++) {
        n = num*i;

        cout<<num<<"*"<<i<<"="<<n<<endl;
    }

    return 0;
}
```

Output :



The screenshot shows a Windows command prompt window with the title bar 'C:\Users\Student\Documents' and standard window controls. The program has executed, displaying the prompt 'Enter any number: 5'. Below this, it prints the multiplication table for 5, from 5*1 to 5*10. The output is as follows:

```
Enter any number: 5
5*1=5
5*2=10
5*3=15
5*4=20
5*5=25
5*6=30
5*7=35
5*8=40
5*9=45
5*10=50

-----
Process exited after 2.297 seconds with return value 0
Press any key to continue . . . |
```


EXPERIMENT – 09

Objective : Write a program to perform size of any datatypes members.

Source code :

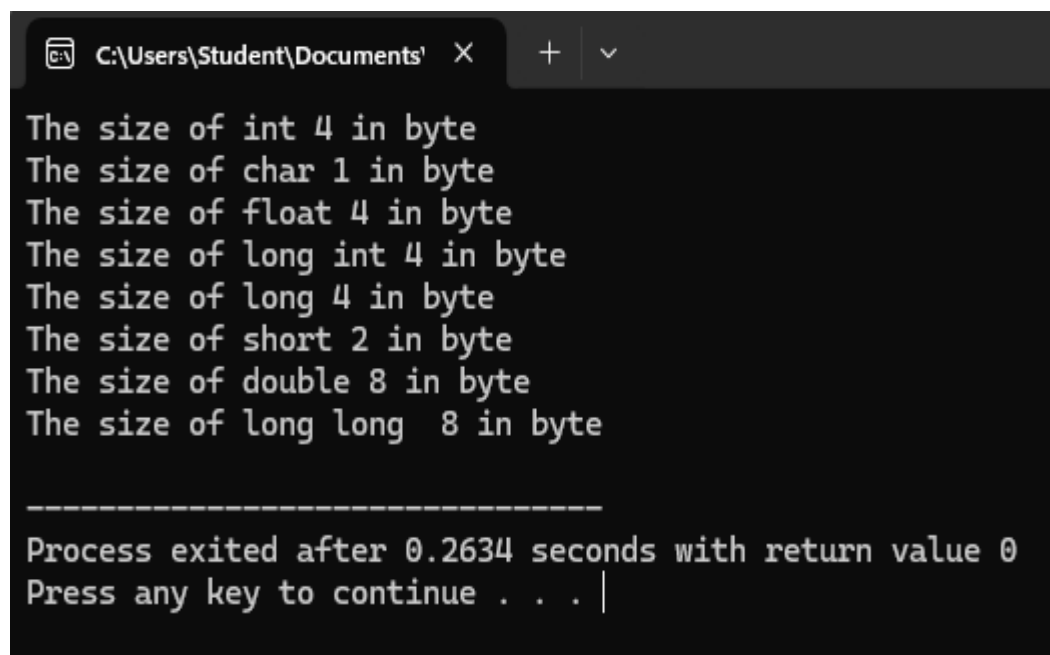
```
#include<iostream>

using namespace std;

int main()
{
    cout<<"The size of int "<<sizeof(int)<<" in byte"<<endl;
    cout<<"The size of char "<<sizeof(char)<<" in byte"<<endl;
    cout<<"The size of float "<<sizeof(float)<<" in byte"<<endl;
    cout<<"The size of long int "<<sizeof(long int)<<" in byte"<<endl;
    cout<<"The size of long "<<sizeof(long)<<" in byte"<<endl;
    cout<<"The size of short "<<sizeof(short)<<" in byte"<<endl;
    cout<<"The size of double "<<sizeof(double)<<" in byte"<<endl;
    cout<<"The size of long long "<<sizeof(long long)<<" in byte"<<endl;

    return 0;
}
```

Output :



```
C:\Users\Student\Documents' X + v

The size of int 4 in byte
The size of char 1 in byte
The size of float 4 in byte
The size of long int 4 in byte
The size of long 4 in byte
The size of short 2 in byte
The size of double 8 in byte
The size of long long 8 in byte

-----
Process exited after 0.2634 seconds with return value 0
Press any key to continue . . . |
```

EXPERIMENT – 10

Objective : Write a program to perform using class and object.

Source code :

```
#include<iostream>

using namespace std;

class A {
    public: int a, b;
    public :
        void input() {
            cout<<"Enter two number: ";
            cin>>a;
            cin>>b;
        }
        void operation() {
            cout<<"sum= "<<a+b<<endl;
            cout<<"sub="<<a-b<<endl;
            cout<<"multi= "<<a*b<<endl;
            cout<<"div= "<<a/b<<endl;
            cout<<"mod=" <<a%b<<endl;
        }
};

int main() {
    A obj;
    obj.input();
    obj.operation();
}
```

Output :

```
C:\Users\Student\Documents' X + v
Enter two number: 50
40
sum= 90
sub=10
multi= 2000
div= 1
mod=10

-----
Process exited after 9.024 seconds with return value 0
Press any key to continue . . . |
```